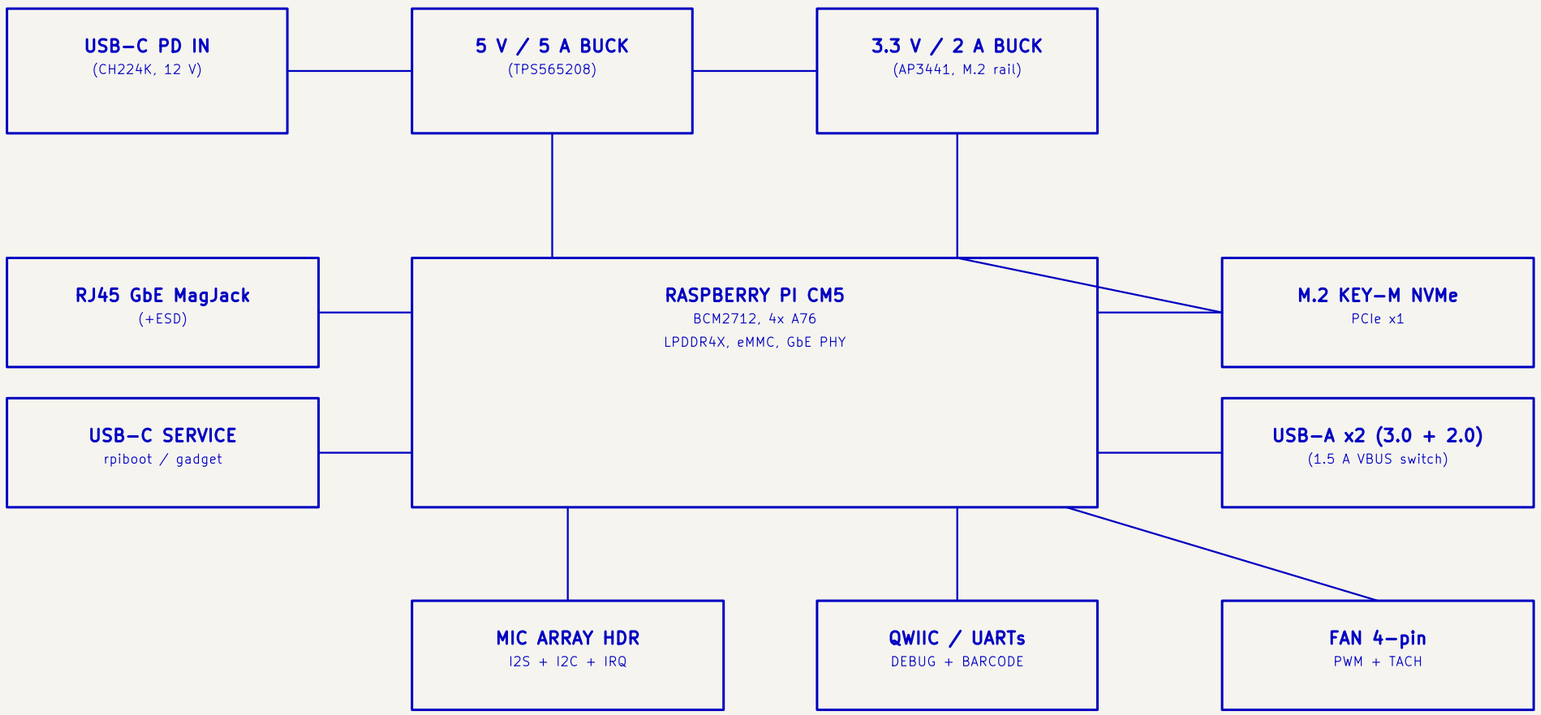
CM5 Edge–Audio Carrier – System Block Diagram



Hierarchical sheets

Power

File: power.kicad_sch

Power: USB–C PD Input, 5 V and 3.3 V Bucks

Compute

File: compute.kicad_sch

Raspberry Pi Compute Module 5

HighSpeed

File: highspeed.kicad_sch

High–Speed IO: PCIe NVMe, USB 3.0, GbE, USB–C Service

PeriphIO

File: periph_io.kicad_sch

Peripheral IO: Mic–Array I2S, UARTs, I2C, Fan

DESIGN NOTES

1. Input: USB–C PD sink requests 12 V / 3 A (36 W). Buck EN divider (100k/15k) holds the 5 V rail off below ~9 V, so a non–PD 5 V source can never back–feed the CM5 at low voltage.
2. CM5 CC1/CC2 left open: set usb_max_current_enable=1 in config.txt to allow 1.6 A on the USB–A ports.
3. PCIe x1 (Gen2 default, Gen3 via dtparam=pciex1_gen=3): 85 ohm diff, length–matched P/N within 0.1 mm. TX AC–coupling capacitors are on the CM5; RX caps are on the NVMe module.
4. USB 3.0 SuperSpeed: 90 ohm diff; GbE MDI: 100 ohm diff; all on L1 referenced to solid GND on L2.
5. ESD: TPD4EUSB30 on every GbE pair, USBLC6–2 on service USB–C, SMAJ15A on VBUS (USB–A per RPi reference).
6. Stackup: JLCPCB JLC04161H–7628, 1.6 mm, L1 SIG / L2 GND / L3 PWR (+5V, +3V3_M2) / L4 SIG.
7. Boot: hold nRPIBOOT while plugging USB–C SERVICE into a host to flash eMMC with rpiboot.

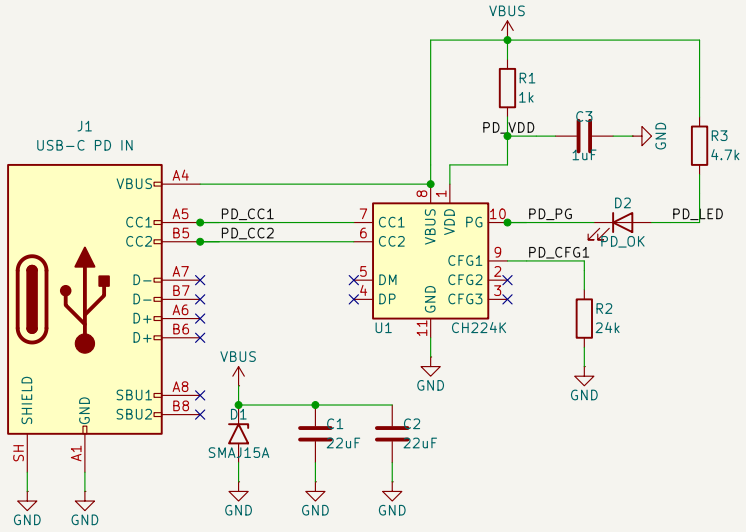
Sheet: Root
Raspberry Pi CM5 carrier for edge–AI audio
Hector Jimenez

Sheet: /
File: CM5_Edge_Carrier.kicad_sch

Title: CM5 Edge–Audio Carrier

Size: A3	Date: 2026–10–21	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/5

USB-C PD SINK (CH224K requests 12 V / 3 A)



CFG1 = 24k -> 12 V PD request (CH224K)

PG low = contract OK -> PD_OK LED on

SMAJ15A clamps hot-plug / cable ringing

Buck UVLO: EN = 1.2 V x (1 + 100k/15k) ~ 9.2 V

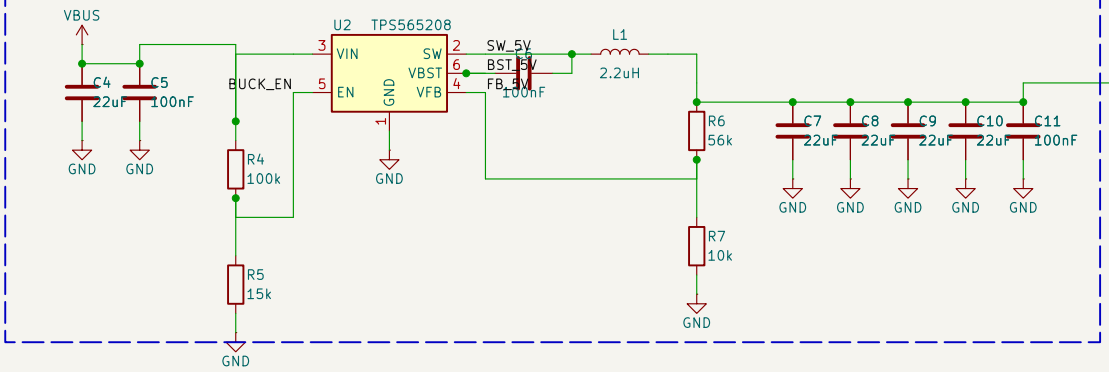
-> +5V stays off if PD falls back to 5 V

Vout = 0.76 V x (1 + 56k/10k) = 5.02 V

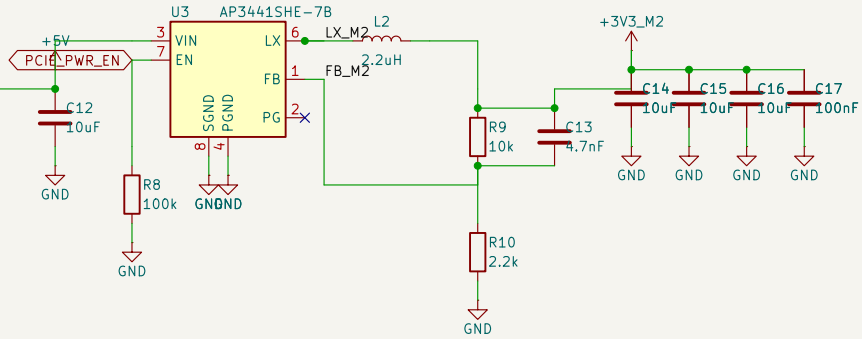
Vout(M.2) = 0.6 V x (1 + 10k/2.2k) = 3.33 V

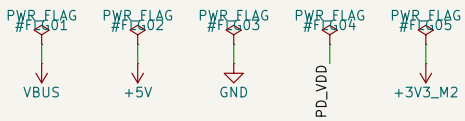
Load: CM5 3.0 A + USB 1.5 A + fan 0.2 A = 4.7 A max

5 V / 5 A SYNCHRONOUS BUCK (TPS565208)



M.2 3.3 V / 2 A BUCK (AP3441, EN = PCIE_PWR_EN)



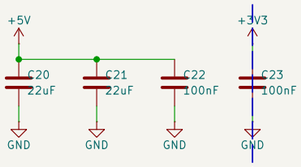


Sheet: Power
Raspberry Pi CM5 carrier for edge-AI audio
Hector Jimenez
Sheet: /Power/
File: power.kicad_sch

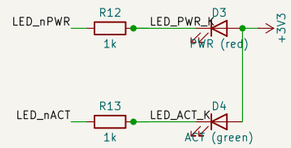
Title: Power: USB-C PD Input, 5 V and 3.3 V Bucks

Size: A4	Date: 2026-10-21	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 2/5

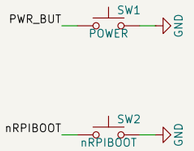
BULK DECOUPLING



STATUS LEDs



USER BUTTONS



nRPIBOOT: hold while powering to enter USB boot;

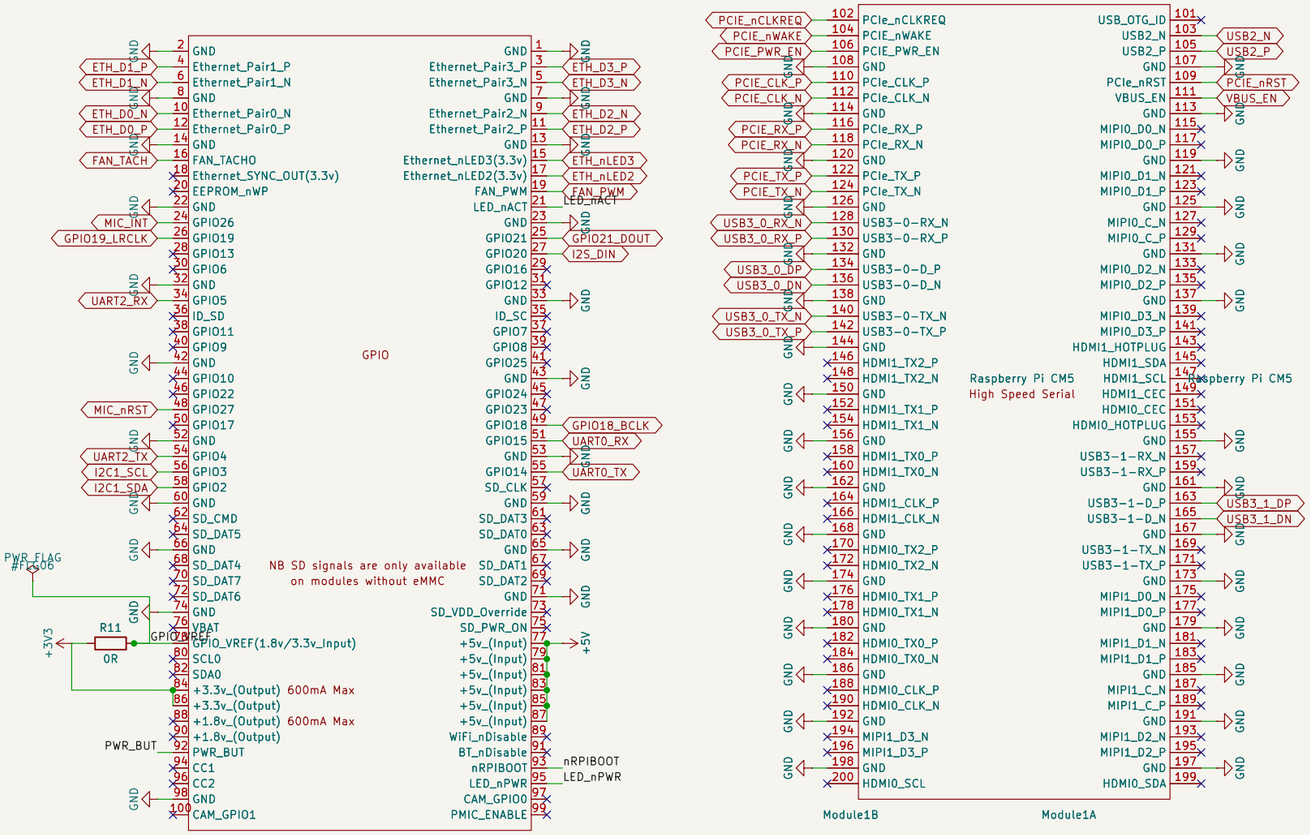
flash eMMC from a host via USB-C SERVICE (rpiboot).

PWR_BUT: short press = wake / clean shutdown.

GPIO bank powered at 3.3 V (GPIO_VREF = +3V3).

Unused: HDMI0/1, MIPI0/1, SD (eMMC SKU), CAM_GPIO,

PMIC_ENABLE, EEPROM_nWP, WiFi/BT_nDisable, VBAT.



Sheet: Compute

Raspberry Pi CM5 carrier for edge-AI audio

Hector Jimenez

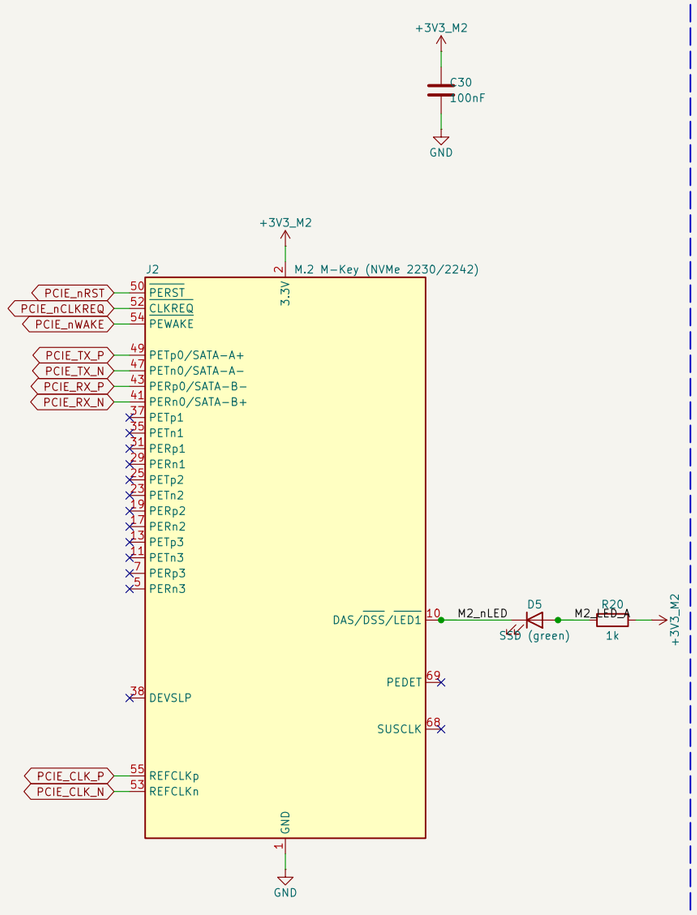
Sheet: /Compute/

File: compute.kicad_sch

Title: Raspberry Pi Compute Module 5

Size: A3	Date: 2026-10-21	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 3/5

M.2 KEY-M NVMe (PCIe x1, 85 ohm diff)



CM5 provides TX AC-coupling caps; NVMe module provides RX caps

USB HOST: 1x USB 3.0 + 1x USB 2.0 (90 ohm diff)

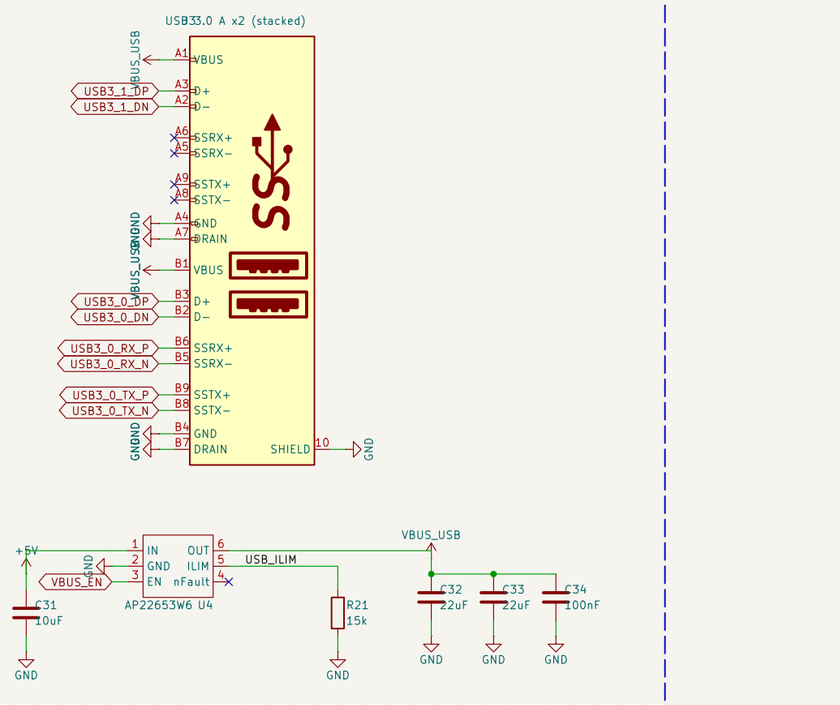
VBUS: AP22653 switch, ILIM = 15k -> ~1.5 A shared; EN = CM5 VBUS_EN

Upper port: USB 3.0 (CM5 USB3-0). Lower port: USB 2.0 (CM5 USB3-1 D+/D-);

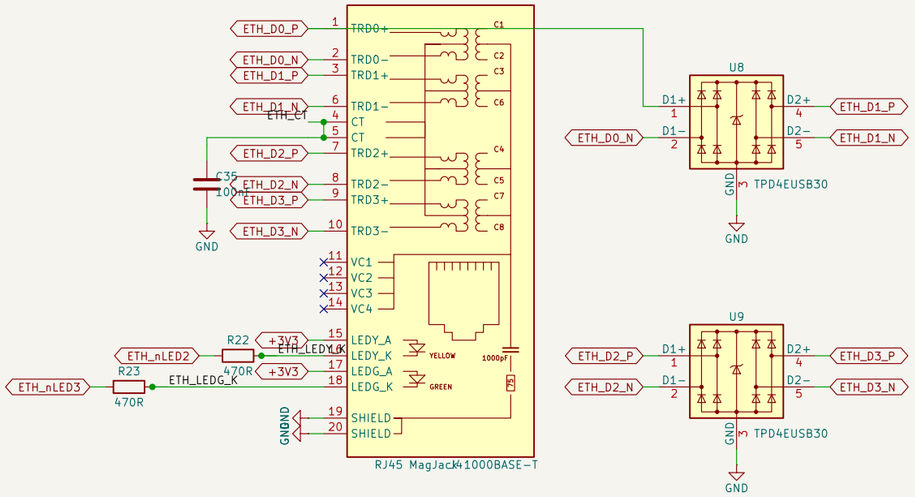
lower-port SuperSpeed contacts left open - routing trade-off, see docs.

USB-A ports carry no ESD array, matching the Raspberry Pi CM5 IO reference:

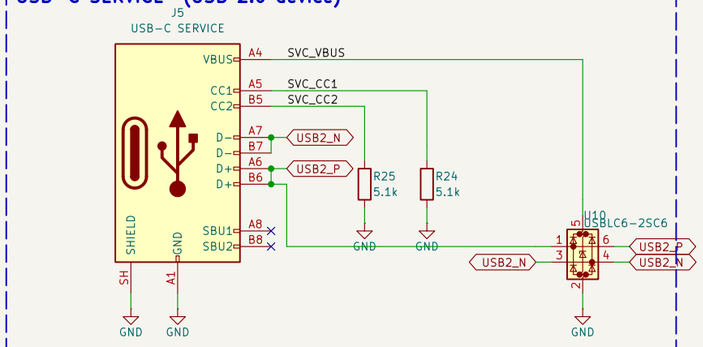
keeps the 5 Gb/s pair stub- and via-free (grounded shell + CM5 PHY protection).



GIGABIT ETHERNET (MagJack + ESD, 100 ohm diff)



USB-C SERVICE (USB 2.0 device)



rpiboot flashing / USB gadget console; CC = 5.1k Rd

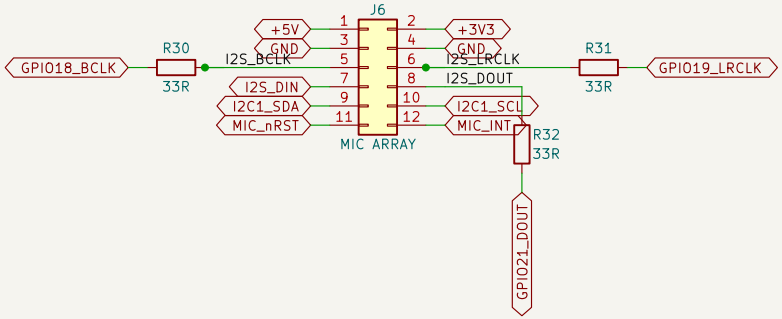
Sheet: HighSpeed
Raspberry Pi CM5 carrier for edge-AI audio
Hector Jimenez

Sheet: /HighSpeed/
File: highspeed.kicad_sch

Title: High-Speed IO: PCIe NVMe, USB 3.0, GbE, USB-C Service

Size: A3	Date: 2026-10-21	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 4/5

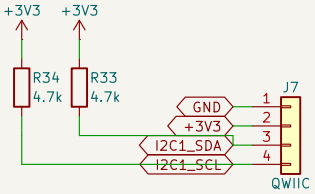
MIC–ARRAY LINK (to XVF3800 6–mic board)



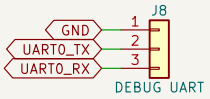
33R source termination on CM5–driven I2S clocks/data

I2C1 pulled up once (4.7k) for header + Qwiic

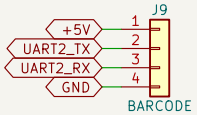
QWIIC I2C (3.3 V)



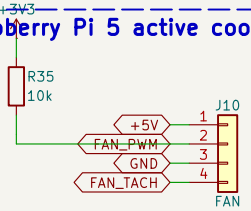
DEBUG CONSOLE (UART0, 3.3 V)



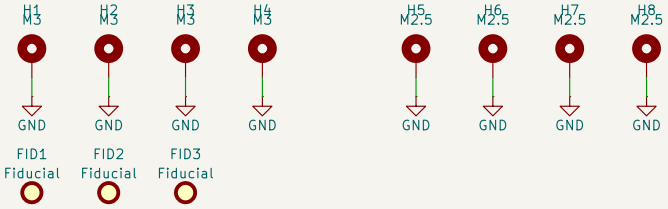
BARCODE SCANNER (UART2, 5 V supply / 3.3 V logic)



FAN (Raspberry Pi 5 active cooler pinout)



MECHANICAL (H1–H4 board corners, H5–H8 CM5 standoffs)



Sheet: PeriphIO
Raspberry Pi CM5 carrier for edge–AI audio
Hector Jimenez
Sheet: /PeriphIO/
File: periph_io.kicad_sch

Title: Peripheral IO: Mic–Array I2S, UARTs, I2C, Fan

Size: A4	Date: 2026–10–21	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 5/5